

Dawoud ROGER

Physics Student

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OBJECTIVES

- Master 2 iCFP
- PhD on Field Theories

EDUCATION

2025 – 2026 | Licence Physics (3rd Year)

Université Paris-Saclay, Orsay
Ranking (first semester) : 2/98
Electives: Advanced Quantum Mechanics, Special Relativity...

2024 – 2025 | Licence Physics (2nd Year)

Université de Bordeaux, Talence
Ranking : 2/54
Elective : Introduction to astronomy

2022 – 2024 | L1-L2 Physics-Chemistry

Université d'Angers
Physics & Chemistry undergraduate studies.

SKILLS

SCIENTIFIC COMPUTING

- 🐍 Python
- 🔗 C
- 📐 LaTeX
- 🐧 Linux Environment

SELF-STUDY

- *Lie Algebras in Particle Physics* (H. Georgi)
- *The Classical Theory of Fields* (L. Landau)
- *The Quantum Theory of Fields I* (S. Weinberg)

LANGUAGES

French: Native
Arabic: Native
English: C1/C2 (Linguaskill)

HOBBIES

Astrophotography
Reading

PROFILE

Passionate about fundamental physics, I made the strategic choice to consolidate my foundations during my second year at Bordeaux, ranking 2nd. Currently in my third year at Paris-Saclay, I am developing expertise in mathematical physics (Lie Algebras, Quantum Field Theory) and numerical simulation during my leisure time. My goal is to join a Master's program to contribute to research in **Theoretical High Energy Physics**.

ACADEMIC PROJECTS

Upcoming Research Internship : Angular Analysis of Λ_b baryon Decay at LHCb

May 2026 – June 2026

IJClab (CNRS) – Orsay

- **Goals :** To be discussed in an *upcoming meeting* with supervisors Marie-Hélène Schune and Nathan Heatley.

Numerical Simulation (Quantum Mech.)

Sept. 2025 – Dec. 2025

Academic Project – Université Paris-Saclay

- **Goal :** Numerical resolution of the *time-dependent* Schrödinger equation in 1D.
- Implementation of the Crank-Nicolson algorithm to ensure stability, utilizing sparse matrix methods.
- Simulations: Harmonic oscillator, Infinite well, and *Quantum Bouncing Neutron*.
- Analysis of wave packet spreading and comparison with analytical solutions.

Research Internship: Gravity Waves

May 2025 – July 2025

Laboratoire d'Ondes et Matière d'Aquitaine (CNRS) – Talence

- **Topic :** Generation of plane surface waves at the air-water interface in the deep-water regime.
- Setup of an experimental design for the Background Oriented Schlieren technique.
- Development of a Python code to visualize and reconstruct the height field $h(x, t)$ from optical distortions.

Bibliography project : Stokes Parameters

Feb. 2025 – May 2025

Academic Project – Université de Bordeaux

- Theoretical study of the Stokes Parameters and their applications in different fields of Physics such as Astrophysics.

Bibliography project : Quantum Computers

Feb. 2024 – May 2024

Academic Project – Université d'Angers

- Theoretical study of quantum entanglement and its applications to quantum computing.